INCH-POUND

MIL-S-13303G(AR)
AMENDMENT 3
30 MAY 1989
SUPERSEDING
AMENDMENT 2
12 March 1988

MILITARY SPECIFICATION

SIGNALS, SMOKE, GROUND, HAND HELD
PARACHUTE, GREEN SMOKE, M128A1,
RED SMOKE, M129A1, AND YELLOW SMOKE, M194
PARTS AND LOADING, ASSEMBLING AND PACKING

This Amendment forms a part of Military Specification MIL-S-13303G(AR), dated 14 September 1984, and is approved for use within the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

3.3 Delete paragraph in its entirety and substitute the following:

"3.3 Cracks. The tube casing shall be free of cracks under visual inspection after hydrostatic testing."

PAGE 9

* Delete from Table I - First Article Inspection:

"Functioning, minus 65°F, 16(a), 3.4.5, 4.5.2.3.2"

PAGE 13

Delete Major 102, add:

"Critical 1 Delay Charge Missing 100% 3.2 Visual"

Change Major 101 from "Igniter composition above flush" to "Delay Composition above flush or more than max. below flush,"

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PAGE 19

Add:

"Critical 2 Overall Density Test 100% 3.2 4.5.5.1"

Notes: "a. Grain shall be assembled and loaded by same manufacturing methods as intended for production.

b. 25 samples to be tested: accept 0,reject 1.

c. 150 grains to be assembled and tested in 50 signals: accept 0, reject 1."

Delete Major 103

PAGE 25

4.4.2.15: Delete in its entirety and substitute the included page.

PAGE 27

4.4.2.17: Delete defects in their entirety and substitute the following:

"Critical

1	Hydrostatic pressure test	100% (Note 1)	3.2	4.5.6
2	Presence of cracks	100%	3.2	Visual

Major

101	Perpendicularity of ends	0.40%	3.2	Gage
102	Total length, max.	0.40%	3.2	Gage
103	Inside diameter at ends, max.	0.40%	3.2	Gage

Minor

201	Poor workmanship	1.00%	3.8	Visual

Note 1 - Tubing stock used to manufacture the casing tube shall be 100% hydrostatically tested as specified in Para. 4.5.6. All remaining classified defects shall be checked on the finished item".

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.15	Tail Assembly				8797947
					NEXT HIGHER ASSEMBLY 8797996/9255782
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Tail fin missing, cracked or inverted		100%	3.2	Visual
Major					
101	Length, max.		0.40%	3.2	Gage
102	Location of tail vanes with diameter and each other including maximum vane extension from centerline		0.40%	3.2	Gage
103	Inside diameter of upper tail ring, min.		0.40%	3.2	Gage
104	Perpendicularity of ends with diameter		0.40%	3.2	Gage
105	Position of rib inside diameter with upper tail ring diameter		0.40%	3.2	Gage
106	Weld missing		0.40%	3.2	Visual
107	Weld flash excessive to extent that function will be impaired		0.40%	3.2	Visual
108	Weld peel test	(a)	--	3.2	4.5.10
109	Hardness test (See Dwg. 8797951)	(b)	--	3.2	4.5.8
Minor					
201	Clearance between lower tail ring and tail vane		0.65%	3.2	Gage
202	Poor workmanship		1.0%	3.8	Visual

NOTES: (a) 32 samples/lot, 1 upper tail ring weld and 1 lower tail ring weld per assembly (64 weld total); (b) Twenty (20) tail vanes shall be selected from each heat treated batch for hardness testing.

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Add to Major 102:

"or improperly attached to chain bead assembly, when applicable."

PAGE 31

4.4.2.21 Delete Major 101 in its entirety and substitute:

"Major 101 Component missing or damaged to extent that function may be impaired or chain bead assembly improperly fastened to smoke assembly, 0.40%, 3.2, Visual"

PAGE 32

Add:

"Critical 2, Initiating Charge Missing, 100%, 3.2, Visual"

Major 103, delete in its entirety and substitute: "Major 103, Initiating Charge Over Max. (see note), 0.40%, 3.2 Balance "

PAGE 36

Delete Major 103

PAGE 38

* 4.4.2.29, Add new Classification of Defects for Signal Body (Dwg. 8797963) included.

PAGE 39

4.4.3.1 Change from "Fiberboard" to "Hardboard"

Change from "Delay Composition (see 3.3)" to "Delay Composition (9251412)"

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.29	Signal Body		8797963
			NEXT HIGHER ASSEMBLY 8797920

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Small inside diameter		0.40%	3.2	Gage
Minor					
201	Large outside diameter, max		0.65%	3.2	Gage
202	Large inside diameter, min		0.65%	3.2	Gage
203	Overall length		0.65%	3.2	Gage
204	Poor workmanship		1.0%	3.8	Visual

NOTES:

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* 4.4.3.2, Functioning. Change NOTE to the following:

"NOTE: The functioning test shall consist of three (3) distinct phases: Hot temperature - fixture fired; plus 70 degrees - fixture fired; and cold temperature - fixture fired Reduced sampling will be authorized only when three (3) consecutive lots have met the acceptance criteria specified in all three (3) phases. Compliance with the criteria in any one phase will not result in reduced sampling for that phase."

Add to Table III

"Cork sealing disc in end of barrel loose Major (see label dwgs. 8797931-2, -3 and -10)"

4.4.3.2.1.1, Line three Change from "twenty-five (25) to "fifty (50)"

4.4.3.2.1.2, Line three Change from "ten (10) to "twenty (20)"

4.4.3.2.2.1, Line four Change from "one hundred and twenty-five (125)" to "one hundred (100)"

Line seven Change from "eight (8)" to "six (6)"

PAGE 41

4.4.3.2.3.1.1, Line three Change from "thirty-two (32)" to "fifty (50)"

Line four Delete "The lot shall ....(see Table III)." and substitute "The lot shall be rejected if, during the test a critical defect occurs, the average angle from vertical exceeds the requirements, or if four (4) or more assemblies exhibit any of the remaining major defects listed in Table III. (See note after the following paragraph)".

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4.4.3.2.3.1.2, Line three Change from "ten (10) to "thirty-two (32)"

Line four Delete in its entirety and substitute: "If any critical defect occurs, the lot shall be rejected." to "The lot shall be rejected if during the test, a critical defect occurs, the average angle from the vertical exceeds the requirement or if three (3) or more assemblies exhibit any of the remaining major defects listed in Table III. (See note).

Note: "A smoke assembly which ignites and burns normally but subsequently fails to meet minimum burning time requirements due to ground impact, will not be classified defective. Firing records will be annotated accordingly."

* 4.4.3.2.3.2, Hand fired: Delete in its entirety.

PAGE 42

Delete paragraphs 4.4.3.3, 4.4.3.3.1, 4.4.3.3.2 and 4.4.3.4

Add new paragraph

"4.4.3.3. Propellant ballistic test - Critical defect. 50 signals per propellant sub-lot of loose propellant composition are to be ballistically tested in fully loaded signals at hot temperature in accordance with 4.5.2.1 abd 4.5.11. Upon successfully meeting the requirements (see para. 3.4.1) the propellant composition used and the method of manufacture of the propellant assembly shall be considered adequate for production. All production will then utilize the spacer, propellant part no. 8797936 in the aft end and the slotted spacer propellant part no. 9328587 in the forward (delay assembly end) section of the tube, casing, part no. 8797921.

PAGE 43

Add:

"4.5.1.5 Propellant assembly, propellant grain. The moisture content of the propellant assembly shall be determined in accordance with the prcedure specified in MIL-P-223 or an approved alternate."

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Add the following note after paragraphs 4.5.2.1, 4.5.2.2, 4.5.2.3.1, and 4.5.2.3.2

"Note: "Samples must be fired within three (3) minutes of removal from the conditioning chamber."

* 4.5.2.3.2, Hand fired: Delete in its entirety.

4.5.6 Delete in its entirety and substitute the following:

"4.5.6 Hydrostatic test of casing tube. In lieu of testing each individual casing tube, the tubing stock, which is used to manufacture the casing tube, shall be 100% tested in a hydrostatic test device (see 4.4.4). The minimum pressure shall be applied for not less than 5 seconds. The tubing shall be observed for evidence of failure to comply with the requirements. Any full length of tubing that bursts, cracks, or shows evidence of distortion shall be classified as defective and removed from production".

PAGE 46

4.5.8 Hardness of tail vane. Add to the end of paragraph:

"If any tail vane fails to comply with the requirement, the heat treated batch represented by the sample shall be rejected".

4.5.11 Delete in its entirety and substitute the following:

"4.5.11 Propellant ballistic test. The signal shall use propellant spacer part no. 8797936, in the forward and aft end while following the test procedure of 4.5.2.1. Simulated or dummy payloads may be used."

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* 6.5, Hand firing: Delete in its entirety.

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